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Calibrating a proportional hazards model with time-correlated covariates: a case study in probability of default modelling for credit risk analysis

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Proportional Hazards Models with an exponential duration model encounter a parameter identification problem in case one or more time-dependent covariates are correlated with the time component of the duration formula. Conventional approaches to modelling time-dependent covariates no longer apply, and a new framework is required. A case study is considered from the field of credit risk modelling to propose a framework that solves the parameter identification problem. Specifically, the effect of credit maturities on the default behaviour of a loan portfolio are often not analysed thoroughly, nor is this a regulatory requirement. However, in most portfolios, there can exist a strong negative correlation between the maturity of a loan and the creditworthiness of the customer. This may lead to spurious conclusions when analysing the default behaviour of the portfolio. In this paper, an example – a selected wholesale Low-Default Portfolio (LDP) at ING – of such a correlation is presented and how this impacts the calculation of a Long-Run Average Default Rate (LRADR) estimate, which is used for calibrating regulatory Probability of Default (PD) estimates. A rigorous mathematical framework based on a proportional hazard rate model, where time is correlated with the scale component of the baseline hazard, is introduced to simulate default patterns comparable to the real-world LDP, which also explains how to parametrise such patterns. Finally, several estimation methods for the LRADR estimate given a correlated time component with a credit ranking function are provided and evaluated.

Keywords: Proportional hazard model; Identification problem; Cox regression; Probability of default curve fitting; Credit risk modelling; Survival analysis; Long run average default rate

Key message

- It is expected that for most credit portfolios the worse the creditworthiness of a lender the less likely the lender will have credits with long maturities.
- A proportional hazard rate model is a good mathematical framework to account for negative correlation between the maturity of a loan and the creditworthiness of a lender.
- A proportional hazard rate model with a significant correlation between the time component and one or

more covariates of the scale component cannot be estimated with conventional methods, but rather with a constrained Maximum Likelihood technique.

1. Introduction

With the introduction of Basel II, financial institutions globally are given guidance on how to calculate a capital short-fall to ensure they are sufficiently capitalised. The level of Tier 1 (Equity) capitalisation is determined by the value of the institution's Risk Weighted Assets (RWA). The RWA formula is provided by the regulator and comprises of several risk parameters, such as the Probability of Default (PD) and the

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Effective Maturity (M) of a credit facility. Should the institution apply for an Internal Rating Based (IRB) approach, the PD parameter is quantified via an internally developed credit risk model and is expected to estimate the Long Run Average Default Rate (LRADR) given the Observed Default Date (ODR) of a portfolio. Note that the LRADR of the ODR is required such that the PD estimate represents a Through-the-Cycle (TtC) constant and does not fluctuate based on changes in systemic risk over time (see EBA 2017, which provide crucial definitions for risk parameter estimation).

Another component of RWA computation is the maturity adjustment, which enters the formula linearly (see Basel Committee on Banking Supervision 2005). This maturity adjustment, however, is a non-linear function of both the PD estimate and M. Thus, the interaction between these parameters has a non-linear impact on RWA, and is highest when both PD and M are low. While the Basel II RWA formula foresees a maturity adjustment in case of IRB, there is no PD adjustment given short-term maturities, which can lead to counterintuitive relationships between an institution's rating grades and their ODRs. In other words, the maturity adjustment only captures the increase in credit, which is also usually translated into a risk premium for pricing loans. However, this does not capture the reduced default risk in case of short-term maturities.

When quantifying the probability of a default event occurring within a given time interval, often this event is considered to follow a Poisson process. This allows us to model the default event as random, with its likelihood dependent on the observed time period and some hazard rate or default event intensity (Chatterjee 2015). Nonetheless, the RWA formula fixes the observed time period to one year. When a loan is reported under the IRB approach, the PD estimate predicts the likelihood of a default event occurring within a subsequent 12-month time period conditional on the facility having a performing (non-defaulted) status at the start of the observation period.

In case a PD is required at periods other than the fixed 12-month horizon, credit modellers often perform survival analyses. A classic approach is the Cox Proportional Hazards (CPH) model (Cox 2018). Such a model has a baseline hazard that increases as the time interval increases, and a vector of covariates that scales this baseline hazard exponentially. When using CPH, a credit modeller defines a vector of covariates that explain the difference in default probabilities independently of time. Subsequently, the time component is used to compute a PD across the lifetime of a loan. The resulting vector of PDs can be used to calculate the Expected Credit Loss (ECL) at any time-interval, which is a requirement for the provisions of banks reporting under the International Financial Reporting Standard 9 (IFRS9). Examples of the success of such models can be seen in Narain (2004), Bellotti and Crook (2009), and Figlewski *et al.* (2012). For a more comprehensive overview of survival models used in PD modelling for IFRS9, the reader is encouraged to read the work in Bank and Eder (2022).

While the time dimension is ignored when calibrating IRB PD estimates, for IFRS9 estimates there is a generally-accepted assumption that time is independent of the covariates used to predict PDs across different loans and/or customers. In addition to this, considering the effects of short-maturity

loans less than 12-months on calibrated PD curves is frequently overlooked. This is driven by IRB models using the 12-month horizon fixed by EU regulation, whereas for IFRS9 a 12-month ECL for Stage 1 loans is the industry standard. However, loans with maturities less than 12 months are common in corporate portfolios, since expired credit facilities with short maturities are not necessarily rolled forward. In other words, when a facility has matured, there is no immediate reopening of the same credit line for a new time period that directly follows the previous credit line. This can occur for a number of reasons, but is often due to proper risk management acumen of the business in charge of underwriting new credit. That is, the maturity of a loan is considered an important tool for an effective credit mitigation policy (Basel Committee on Banking Supervision 2001). The existence of this practice is most prominent for creditors in worse rating grades. This makes intuitive sense. Front office management is less likely to issue a long-standing credit line to a creditor with a bad credit rating. As a result, the LRADRs calculated for bad credit ratings are lower than expected, which can lead to a systematic violation of the monotonicity of the LRADR across the credit rating scale. This is problematic for an institution that calibrates their PDs using external default data, as internal LRADRs will not be representative of the default rates collected from external rating agencies. An example of this phenomenon in a selected wholesale Low-Default Portfolio (LDP) at ING is showcased to highlight this fact.

As is shown in this paper, the semi-parametric form of the CPH model leads to optimistic PD estimates if this business practice is commonplace within the institution. The CPH model fails to accurately calibrate under this circumstance because this negative correlation between maturity and credit rating is significant, violating the CPH model's key assumption. Specifically, a CPH model assumes that the duration of an observation is independent of the model's covariates, which, as we've described, is not always the case in loan default prediction. The implication of this is that the conventional Breslow estimator (Breslow 1972, Lin 2007) should not be used for credit risk modelling. It is often proposed to introduce an interaction term between the duration and the time-dependent covariate (Bellera *et al.* 2010), however this does not solve the parameter identification problem for the baseline hazard and the correlated covariate. This paper presents a novel duration adjustment to the CPH model, and estimates this on simulated data via a constrained maximum likelihood technique. This study finds that such an adjustment results in accurate calibration to the true LRADRs where other techniques fail when the CPH model's independence assumption is violated.

The paper is structured as follows. Section 1 introduces how risk management practices can lead to a negative correlation between a facility's short-term maturity and the creditworthiness of the client, with a detailed example of how this impacts the real-world LDP. In Section 2 the counterintuitive relationship between a credit rating and the LRADR – as observed for ING – is replicated by simulating defaults using a CPH model with a time component negatively correlated with the observation's credit rating. The mathematical framework in this section further explains how to estimate the parameters of a proportional hazard function when the time

component of the baseline hazard is not independent of the scaling factor. Finally, Section 3 proposes a method to calculate LRADR estimates adjusted for maturity, such that the estimates predict the 12-month default probability independent of the common factor, as expected by IRB regulation. This section also explains how to use the maximum likelihood estimators of a proportional hazard function from Section 2 to calculate a LRADR estimate. The paper summarises the findings and conclusions in Section 5.

2. Loan maturity and credit risk

There is a need to account for credit maturity in PD estimation and to investigate its impact on LRADR estimates going beyond what current regulation foresees. Specifically, the Basel framework for calculating the short-fall on credit losses elucidates the impact of maturities on credit losses, but not on the PD itself. Consider the RWA formula for sovereign, corporate, and bank exposures provided by regulation is given as:

$$R(PD, LGD, EAD, M) = 12.5 \times g(PD, M) \times LGD \times EAD, \quad (1)$$

where the function $g(PD, M)$ is given as

$$g(PD, M) = \left[\Phi \left(\frac{\Phi^{-1}(PD)}{\sqrt{1-\rho}} + \sqrt{\frac{\rho}{1-\rho}} \Phi^{-1}(0.999) \right) - PD \right] f(PD, M),$$

$$\rho = 0.12 \frac{1 - e^{-50PD}}{1 - e^{-50}} + 0.24 \left(1 - \frac{1 - e^{-50PD}}{1 - e^{-50}} \right),$$

and the maturity adjustment, $f(PD, M)$, is given as

$$f(PD, M) = \frac{1 + (M - 2.5) (0.11852 - 0.05478 \log(PD))^2}{1 - 1.5 (0.11852 - 0.05478 \log(PD))^2}. \quad (2)$$

Here, the key risk parameters are the Probability of Default (PD), Loss Given Default (LGD), and the Exposure at Default (EAD). M is the maturity of the loan and ρ is the asset correlation.[†] As can be seen in equation (1), a facility's RWA depends linearly on LGD and EAD, but non-linearly on PD. The interaction between PD and M through the maturity adjustment in equation (2) also leads to a non-linear effect on the RWA. Specifically, the highest uncertainty with respect to expected credit losses arises for short-term credit products in combination with solvent creditors, and is illustrated in figure 1. However, when the expected value of credit losses is of interest and/or the PD for a fixed time period, the impact of maturity on the LRADR is of importance.

[†] As per the Basel framework, an additional multiplier of 1.25 is applied to the asset correlation for loans to all unregulated financial institutions and regulated financial institutions whose total assets are at least USD100 billion.

2.1. Risk management and credit writing

To ensure uniform comparability of credit risk profiles between different clients and portfolios, financial institutions and credit bureaus define their own internal standardised reference scale. This scale, often referred to as a MasterScale (MS), is a regulatory requirement. Specifically, the EBA Guidelines on PD estimation (EBA 2017) require PD estimates to be assessed by pools of lenders that have a similar idiosyncratic risk profile. This MS is required to be monotonic, either increasing or decreasing. In addition to this, the difference in default risk as one moves from one rating to the next are expected to be significant, and usually represent exponential jumps across the full scale.

Keeping this in mind, consider now the following simple, yet realistic, scenario for wholesale credit departments that are not high-volume oriented. A credit officer is in charge of generating revenue for the organisation by means of selling credit and limiting credit losses by understanding the credit profile of the customer. Since riskier clients usually translate to better margins, i.e. more profit for the organisation, a good trade-off is required between profitability and the credit risk embedded in the portfolio. The credit officer understands that a default event is random and can be avoided by following up closely on the liquidity of the client and the exposure of the bank towards the client. If the liquidity of the client is under stress, the credit officer can stop issuing new credit. If, however, the credit product has a long-term maturity this limits the possibility of the credit officer to intervene quickly.

As part of a good first-line risk management process, it is thus expected that solvent clients with a good credit history are allowed to receive credit products with longer term maturities, while insolvent clients and/or clients with bad credit history will receive credit products with shorter term maturities. For these latter clients, their credit exposure is also expected to be limited. The discontinuity of lending of an organisation's largest lender often triggers a default event, especially if unannounced in advance. This usually occurs because the organisation has limited alternative lending options. Explicitly, it is thus expected that small volume wholesale portfolios have a negative correlation between the creditworthiness of clients and the maximum residual maturity of all credit facilities connected to that client. Moreover, there exists a positive correlation between the creditworthiness of a client and the total sum of credit exposure towards a client. This entanglement of issuance of credit based on a credit office's contemporary knowledge of the customers in a portfolio and the resulting number of defaults observed creates a bias unaccounted for in current regulatory frameworks.

2.2. Real-world portfolio

An example of the effect described above can be seen in figure 2 for a wholesale LDP at ING. The model shown (blue) is the result of a ranking function that allocates observations to ING's internal MS. The LRADR for each rating grade is then computed as the arithmetic average of a 12-month forward-looking default flag. For comparison, the LRADR implied by the external rating grades of S&P for the same portfolio are plotted. Note that the scale implies that creditworthiness of the

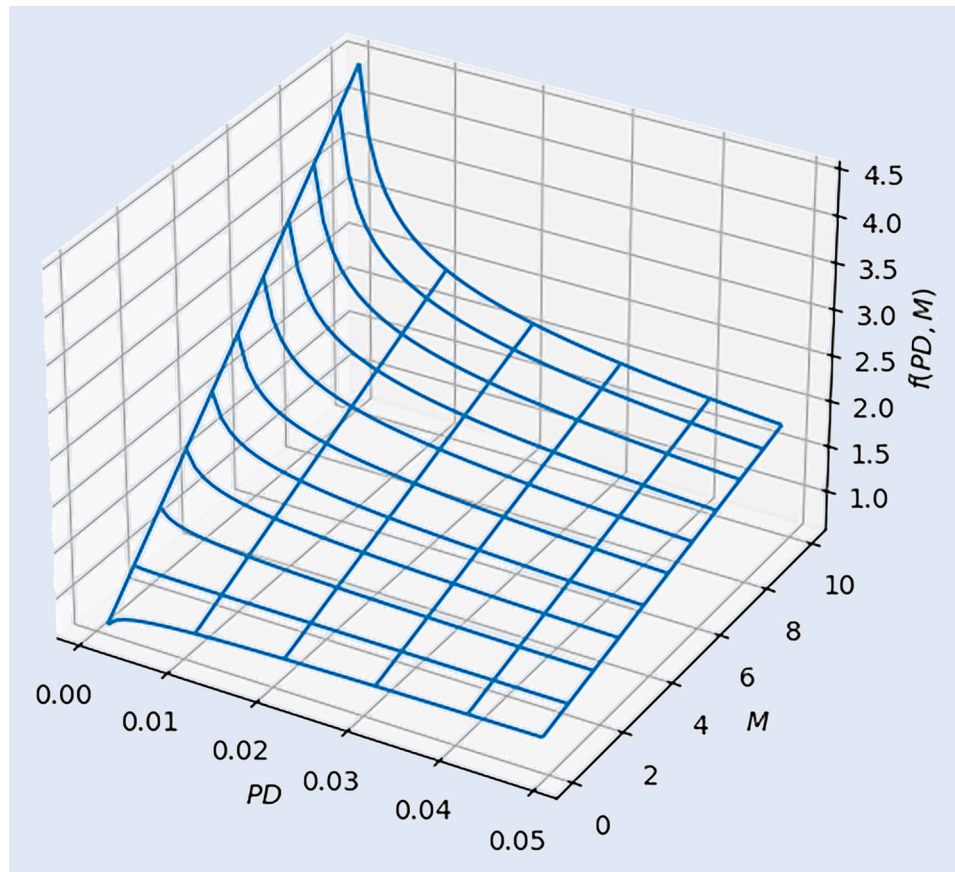


Figure 1. Wire plot of the maturity adjustment in the Basel RWA formula.

obligor decreases as the rating grade increases. For this portfolio, the obligors that have internal and external ratings overlap considerably, and care has been taken to ensure that there is direct comparability between the S&P grade scale and ING's internal MS, thus the issue of representativeness is minor.

It can be clearly seen that the internal default curve is flatter than the one based on the S&P rating scale. Specifically, from rating grade 11 onwards the internal default rate remains flat with only the worst rating grade having a higher LRADR. There are a number of reasons this could occur. An initial thought would be that the ranking function performs poorly, i.e. that it does not discriminate adequately between good and bad loans. However, by reproducing this plot grouping obligors using the external S&P rating the problem persists, as can be seen in figure 3. The default curve based on ING's internal default flag remains flat up until grade 14 where this increases sharply before flattening out again. Hence, this counterintuitive relationship for the internal default curve cannot be explained by the performance of the ranking function.

Given the above argumentation and plots, the empirical correlation between the internal/external rating grade and the maximum maturity of all facilities linked to an obligor is computed. These are computed for the subset of the portfolio where the maximum maturity is either less than one year or longer than one year and are given in table 1.

It is noted that for maturities less than one year there is a negative relationship with the rating grade, that is, as the rating grade of the obligor increases (worse creditworthiness), the maximum maturity shrinks. For the range of maturities

Table 1. Pearson correlation coefficients between maximum maturity of an obligor and the internal/external rating grade.

Maximum maturity ($M_{\max}$)	Internal grade	S&P grade
$M_{\max} > 1$	-0.274	-0.143
$M_{\max} \leq 1$	-0.234	-0.138

larger than one year this negative relationship persists. This implies that as the rating of the obligor decreases (better creditworthiness), the maximum maturity tends to be longer. In the next section, a simulation that enforces this relationship between the rating of an observation and its maturity replicates this effect of a flat default curve.

3. Survival analysis

In this section defaults are simulated via a Poisson process following the architecture of Bender *et al.* (2006), but with an induced correlation between loan duration and loan score. Standard hazard rate models are then fit to show how these fail when important assumptions are violated. Similar to evaluation techniques used to price financial instruments (see, e.g. Hull *et al.* 2005), the hazard rate or default intensity will be used to understand the likelihood of defaulting within a certain time interval. Thereafter, an alternative maximum

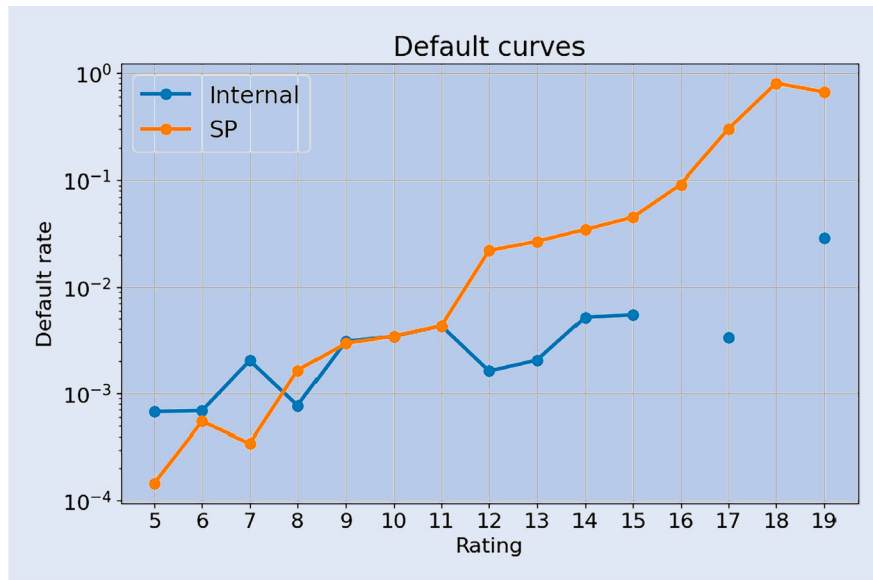


Figure 2. Default curves of a wholesale LDP at ING with its MS as the reference.

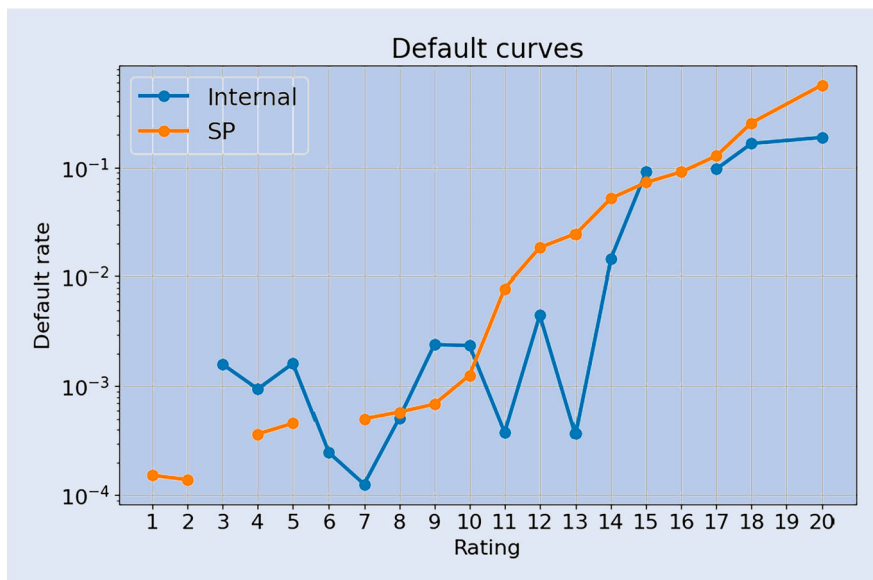


Figure 3. Default curves of a wholesale LDP at ING with the S&P rating as the reference.

likelihood method that explicitly identifies the functional form of the baseline hazard function is introduced and compared to standard methods.

3.1. Simulation

Consider a portfolio of $N = 1000$ facilities, all with varying levels of creditworthiness and maturities. In order to formally write down the intuition of the expected impact of maturity on PD, a few standard assumptions are introduced. These are:

- (1) For any given facility, it should be possible at any given moment in time, t , to verify if it is performing or defaulted within a 12-month forward-looking horizon via the Point-in-Time (PiT) binary state $D(t)$.
- (2) The credit quality, with respect to PD, of a facility is dependent on two factors:

- (a) an idiosyncratic factor, which is given by a discrete score (S) ranging from 1 to 19 representing the inherent creditworthiness of the facility, and;
- (b) a systemic factor (Z_t) representing the underlying credit cycle, which affects all facilities equally.
- (3) Between each disjoint time interval there is full independence between the default or performing state of the facility. In other words, the PiT state of the facility is independent of its previous state.

When calculating LRADR it is not the number of default events that can occur over a given time interval that is of interest. Instead, one predicts at any given future point whether the facility has not yet defaulted, and what the likelihood of the default event would be given the forward-looking time horizon. This binary outcome in discrete time follows a Bernoulli process. However, since at any given moment – and not necessarily at fixed time intervals – the default state

is assessed or predicted, a Poisson process will be assumed instead.

Consider a hazard rate model for the PiT PD, where the hazard rate is a function of the score and systemic factor:

$$PD_{PiT}(t | S, Z_t) = 1 - e^{-\lambda(t|S, Z_t)}. \quad (3)$$

Assuming that the hazard rate is exponentially proportional to the covariates, it can be written as follows:

$$\lambda(t | S, Z_t) = \lambda_0(t) e^{\beta_1 S + \beta_2 Z_t}, \quad (4)$$

where the covariates vary through time t . A Through-the-Cycle (TtC) PD implied by the above model is given by integrating over all values of the systemic factor Z_t . This yields an unconditional PD dependent solely on S :

$$PD_{TtC}(S) = e^{-LRADR(S)}, \quad (5)$$

where the LRADR in this context is equivalent to the time-weighted ODR for score S . For now the loan duration is assumed to be $\Delta t = 1$.

Using equations (3) and (4), default events are simulated over time. Consider the following default event construction:

$$D(t) = \begin{cases} 1, & \text{if } u \leq 1 - e^{-0.00181 \Delta t e^{0.28S - 0.30Z_t + 0.01\varepsilon_1}} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where,

$$\begin{aligned} \Delta t &= \min \{1, \max \{0.01, 1 - 0.06S + 0.08\varepsilon_2\}\}, \\ u &\sim U(0, 1), \\ \varepsilon_1, \varepsilon_2 &\sim \mathcal{N}(0, 1). \end{aligned}$$

In equation (6) the loan duration variable Δt has its units in years, is capped at 1 year, and floored at ~ 3.6 days. Note that the loan duration is shortened when the loan's score worsens. The baseline hazard function in the data generating process is thus also dependent on the idiosyncratic risk factor through the duration variable. Specifically, Δt is dependent on the risk profile of the observation, leading to a dependency of λ_0 on both t and S . The coefficient for S in the loan duration thus captures the negative relationship between creditworthiness and loan maturity described in Section 1. Other aspects of the simulation are given in table 2.

Using the results of the simulation, a MasterScale and its respective LRADR is plotted in figure 4. The LRADR per score S is calculated as the mean of the one year ODRs. The same counterintuitive relationship between S and the simulated LRADR as described in earlier sections is observed. In particular, the MS exhibits an overall increase in LRADR as scores S increase, but then systematically decreases for the highest (worst) S values.

3.2. Cox regressions

Given the simulated ODRs in figure 4, two standard Cox Proportional Hazards models are calibrated to assess how well they handle situations where fundamental assumptions

Table 2. Simulation architecture and underlying data.

Total entities in the portfolio	1000 ¹
Time period (quarterly)	2007:Q1-2025:Q1
Score	Uniform integer ² , $S \in \{1, \dots, 19\}$
Common factor ³	GDP European Union - 27 countries (from 2020) ⁴

Notes: ¹This choice best reflects the typical size of a bank portfolio and is considered sufficient to reach the central tendency of the LRADR per score. ²The assumption of a uniform rating distribution is not realistic, as we expect the tails of the ratings in real portfolios to be less populated than the middle ratings. The sampling error associated with this could also explain non-monotonicity of the LRADR, but as the results of the simulation show, this effect is not strong enough to drive systematic drops in the LRADR for the worst rating grades. ³Change of GDP normalised using the Empirical Cumulative Distribution Function (ECDF). See Oeyen and Celis (2019) for more information on the variable transformation logic. ⁴Data extracted from Eurostat.

are violated. Both assume the baseline hazard as independent of the two covariates, but the second one introduces a third covariate as an interaction term between the idiosyncratic and systemic risk factors. This is done to potentially capture any effects not explained by the two factors alone.

Consider a portfolio with $i = 1, \dots, N$ facilities observed through time $t = 1, \dots, T$ and a general set of p covariates, $x_i = [x_{i,1}, x_{i,2}, \dots, x_{i,p}]^T$. Let the PiT performing or default status for observation i be defined as $D_i \in \{0, 1\}$ and the risk set, $K(s)$, be the subset of facilities still at risk of default at time t_s ,

$$K(t_s) = \{i : D_i = 0 \mid t_s \geq t, \forall t = 1, \dots, T\}.$$

The standard CPH model optimises a partial loglikelihood with respect to the coefficient vector $\beta = [\beta_1, \beta_2, \dots, \beta_p]^T$,

$$L(\beta) = \sum_{i=1}^N D_i \left[x_i^T \beta - \log \left(\sum_{j \in K(t_s)} \exp(x_j^T \beta) \right) \right], \quad (7)$$

producing a maximum likelihood estimate, $\hat{\beta}$. This is then used to get an estimate of the baseline hazard function via the standard Breslow estimator (see, e.g. Lin 2007),

$$\hat{\lambda}_0(t_s) = \sum_{j: t_j \leq t_s} \frac{D_j}{\sum_{i \in K(t_j)} \exp(x_i^T \hat{\beta})}, \quad (8)$$

which relies on the assumption of independence between the baseline hazard function and the covariates, x_i . The proportional hazards assumption is assessed by looking at the scaled Schoenfeld residuals (see Schoenfeld 1982 or Grambsch and

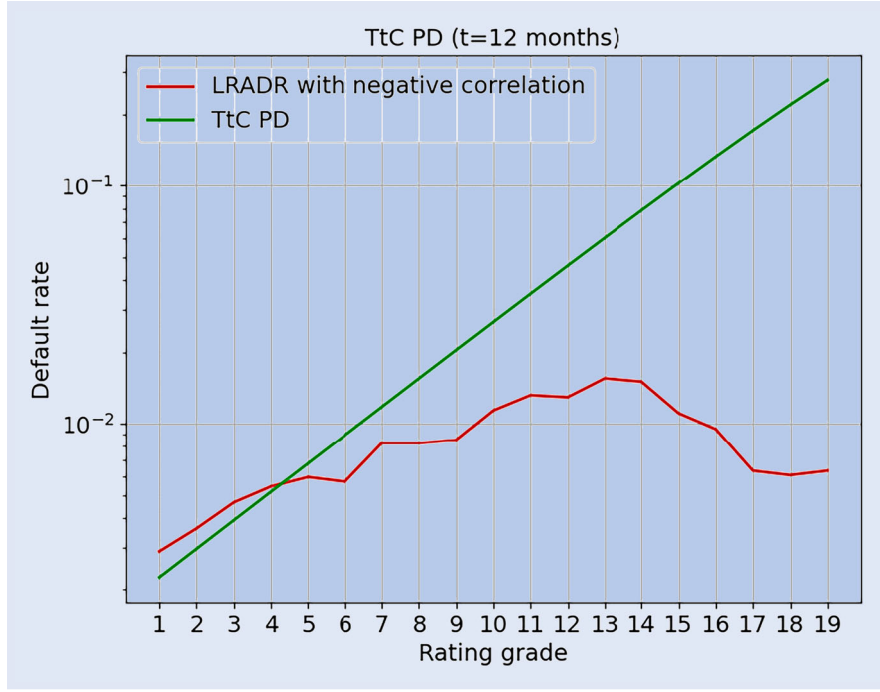


Figure 4. Simulated MasterScale. As one reaches the worst scores, the LRADR systematically drops.

Therneau 1994). Specifically, for an observation i and covariates $k = 1, \dots, p$, the scaled Schoenfeld residual is defined as:

$$r_i = \begin{bmatrix} r_{i,1} \\ r_{i,2} \\ \vdots \\ r_{i,p} \end{bmatrix} = \Sigma^{-1}(\hat{\beta}) \begin{bmatrix} e_{i,1} \\ e_{i,2} \\ \vdots \\ e_{i,p} \end{bmatrix}. \quad (9)$$

Here, $\Sigma^{-1}(\hat{\beta})$ is the inverse of the estimated covariance matrix of the partial score function at each time step, and the vector $e_i = [e_{i,1}, e_{i,2}, \dots, e_{i,p}]^T$ is the difference between the observed covariate value and the expected value of that covariate given the risk set of the model,

$$e_{i,k} = x_{i,k} - \frac{\sum_{j \in K(t_s)} x_{j,k} \exp(x_j^T \hat{\beta})}{\sum_{j \in K(t_s)} \exp(x_j^T \hat{\beta})}. \quad (10)$$

If a trend in the residuals is observed for any of the covariates in the model, the conclusion is that there is a time dependence unaccounted for in the model. The standard CPH model is applied to the simulated ODRs with covariate vector $x_i = [S_i, Z_{i,t}]^T$ with the Schoenfeld residuals computed using both estimated and true coefficients, as shown in figure 5. The test results are also shown for the true coefficients to illustrate the importance of estimating a reliable baseline hazard in the Schoenfeld test.

Figure 5 illustrates the key properties of the covariates concerning their relation to the time component of the duration model. Firstly, for the systemic factor ‘Z’ (Top-Right and Bottom-Right), residuals are flat for both estimated and true coefficients, confirming the proportional hazards assumption holds for the systemic factor. Secondly, for the ‘Score’ variable (Top-Left and Bottom-Left) a significant trend is evident

in residuals for both estimated and true coefficients. This confirms the violation of the proportional hazards assumption for ‘Score’, indicating that the time dependence is an inherent characteristic of the simulated data’s underlying duration model. Finally, the results also illustrate the bias in the Breslow estimator when one or more covariates are correlated with the time component in the duration model.

A common approach to tackle the problem seen in figure 5 is to introduce an interaction term between the duration variable, and the variable violating the proportional hazard assumption (Bellera *et al.* 2010). The result is a time-varying CPH (TVCPH) calibration with covariate vector $x_i = [S_i, Z_{i,t}, S_i \times Z_{i,t}]^T$, which is expected to capture the interaction between score and duration. However, this does not result into reliable parameter estimates as is demonstrated later in this section.

3.3. Alternative maximum likelihood method

A maximum likelihood estimation that explicitly estimates the baseline hazard as a scalar coefficient, the LRADR, multiplied by the duration of the observation itself is implemented and compared to the previous models. For ease of distinction, this model is called the Parametric Baseline Cox (PBC) regression model.

The standard CPH model maximises the partial loglikelihood (Cox 2018) in equation (7), which relies on the assumption that the constant baseline hazard does not influence the location of the function’s optimum. If the assumption of a constant baseline hazard is relaxed, then this is no longer possible and the likelihood must be explicitly defined. The PiT PD enters the binomial loglikelihood in the usual way, with the functional form of the PD matching the functional form given in equation (3). An extra parameter, γ , representing the

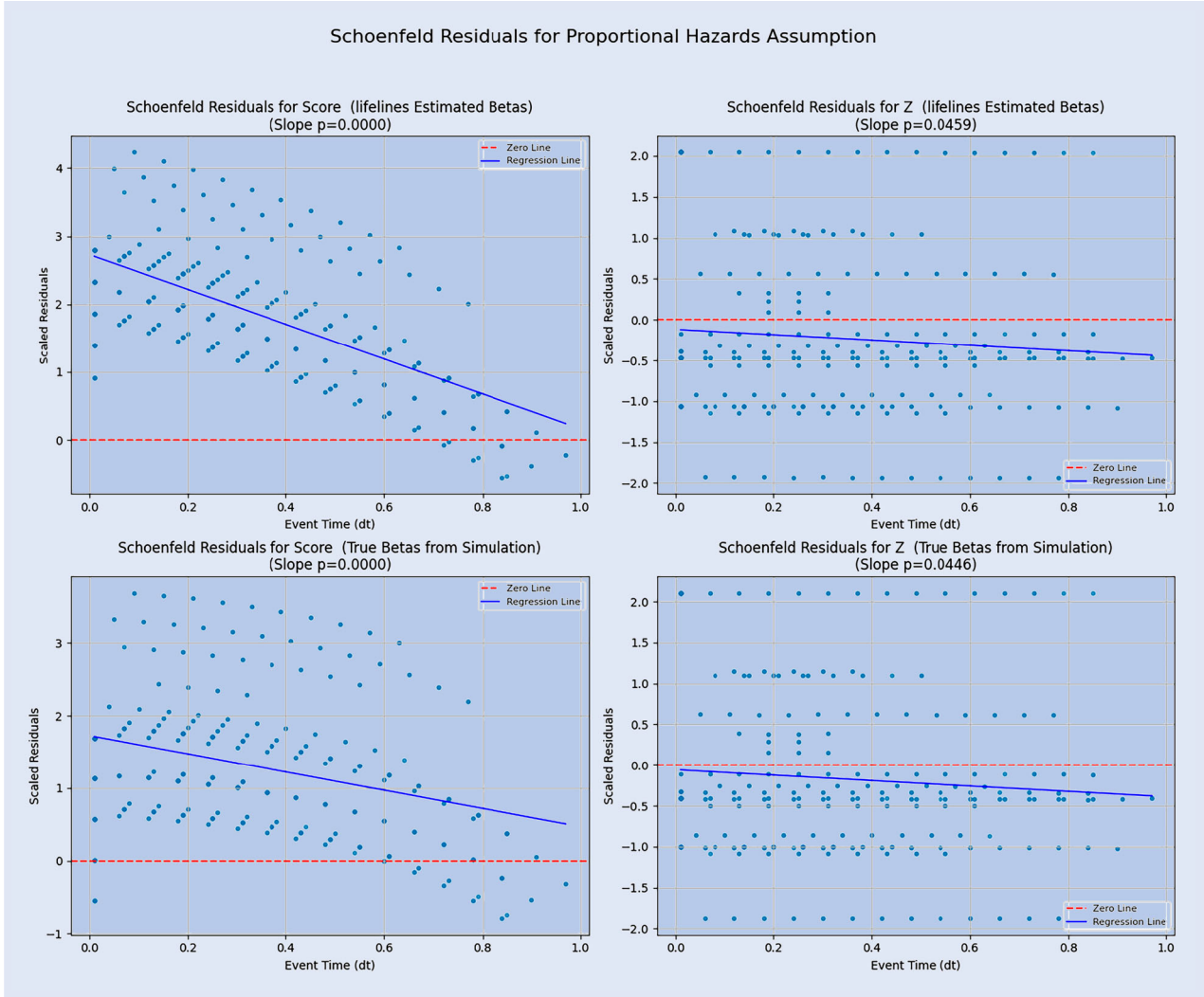


Figure 5. Scaled Schoenfeld residuals for ‘Score’ and ‘Z’ covariates using both estimated and true coefficients. Top-Left: Score (Estimated Betas); Top-Right: Z (Estimated Betas); Bottom-Left: Score (True Betas from Simulation); Bottom-Right: Z (True Betas from Simulation).

TtC PD is to be estimated within the baseline hazard function:

$$\lambda_0 \equiv \gamma \Delta t.$$

Collecting the three parameters into the vector $\vartheta = \{\gamma, \beta_1, \beta_2\}$, the hazard rate is thus parameterised as follows:

$$\lambda_i(\vartheta | S_i, Z_{i,t}) = \gamma \Delta t_i e^{\beta_1 S_i + \beta_2 Z_{i,t}},$$

where each observation has its own unique duration variable and score. An important point is made here: no explicit link is made between the score and loan duration. The constant TtC PD is simply de-coupled from the loan duration variable, which is a known variable in the dataset. This gives the following target function for optimisation:

$$L(\vartheta) = \sum_{i=1}^N D_i \log(1 - e^{-\lambda_i(\vartheta | S_i, Z_{i,t})}) - \sum_{i=1}^N (1 - D_i) \lambda_i(\vartheta | S_i, Z_{i,t}). \quad (11)$$

The above optimisation problem differs from the standard Cox regression as it defines a full loglikelihood instead of

a partial loglikelihood. This fully parametric approach to Cox regression is never done, as an immediate problem of model identification arises. This is, the parameter γ can be brought into the exponent of the hazard function and combined linearly with β_1 and β_2 . As a result, unconstrained optimisation of the above function will not produce robust, consistent estimates. To sidestep this issue, the model’s monotonic relationship between an observation’s PD and score can be leveraged to bound the optimisation problem. The logic is as follows. The TtC PD is by definition lower bounded by zero. Since the exponent of the covariates will always scale the baseline hazard such that it increases, an upper bound to γ can be set to the empirical default rate of the observations with the best scores – or, in our case, the supremum of rating grades ($S = 1$) – and a loan duration of one year or higher ($\Delta t \geq 1$).[†] A higher rating grade may also be chosen in case no default rate can be calculated due to an absence of default observations. This ensures that the optimisation is narrowed down to the most plausible region for the true LRADR. Minimisation of the negative of equation (11) can then be carried

[†] This avoids the scaling of the baseline hazard by the time component.

out by any bounded, constrained optimisation algorithm, such as a truncated Newton method. Bounds on the parameters for β_1 and β_2 are not strictly necessary, however, one may wish to enforce strict positivity of β_1 , as this forces a positive relationship between the PD of an obligor and their score. This may be reversed in case the creditworthiness of customers are expected to improve as the score increases.

3.4. Fitted models

Table 3 shows the calibrated coefficients for all models. The standard CPH model calibrates poorly, which is expected, since the proportional hazard assumption is violated. The TVCPH and PBC models correctly estimate the β_2 coefficient for the systemic factor. However, the addition of an interaction term in the TVCPH model still does not correctly calibrate β_1 and produces a spuriously significant coefficient for the non-existent interaction term. While no explicit link is modelled between score and maturity in the PBC model, its estimation yields a calibrated value much closer to the true value. In addition to this, the truncated Newton method also yields a TtC PD for γ that is close to the true value. This shows that the PBC model captures the effect of a strong positive correlation between maturity and score without needing to explicitly define their relationship or how these interact.

4. Estimating the LRADR

In this section several methods are introduced and back-tested to calculate a LRADR of 12-month ODRs on a discrete ranking interval. Default events are simulated with a baseline hazard that is dependent on the maturity of the facility, as is described in the previous section.

Consider the framework given in Section 2 where a loan duration variable is introduced Δt , which may not necessarily equal 1 year.[†] Under this framework the following information is known to us for all facilities i in the portfolio: all binary states, D_i , and their duration ($\Delta t \in (0, 1]$) at different snapshots ($t = 1, \dots, T$), and their score, which is an integer $S \in \{1, \dots, 19\}$. With this information the LRADR can be calculated in several ways. Four different approaches are explained and calculated on a simulated dataset and their respective LRADR curves compared.

4.1. Classic

This method is most commonly used in IRB banks as the time interval considered is exactly the one year observation window prescribed in regulation, i.e. $\Delta t_i = 1 \forall i$. Note that no correction is foreseen here to account for observations for which the time horizon is below one year. Here, the LRADR

is calculated as the ODR per observation period and score:

$$LRADR_S = \frac{\sum_{t=1}^T \frac{\sum_{i=1}^{N_{S,t}} D_i}{\sum_{i=1}^{N_{S,t}} 1}}{\sum_{t=1}^T 1} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_{i=1}^{N_{S,t}} D_i}{N_{S,t}}, \quad (12)$$

where $N_{S,t}$ is the number of observations with score S at time t . Therefore, the formula provides the central tendency of an observation-weighted default rate.

4.2. Duration

Under this approach the LRADR is computed as a duration-weighted ODR per score:

$$LRADR_S = \frac{\sum_{i=1}^{N_S} \Delta t_i D_{i,S}}{\sum_{i=1}^{N_S} \Delta t_i}. \quad (13)$$

Equation (13) provides a LRADR for the expected maturity of the portfolio conditional on the given score, i.e. $\mathbb{E}[\Delta t | S]$. In case the portfolio contains legal entities that have several facilities with the lending institution and the score and default event is determined on customer level, only the facility with the highest maturity is relevant. This reflects the ability of the organisation to intervene timely prior to a customer default event. In other words, only one Δt should be considered per legal entity.

Nonetheless, the LRADRs in this example are expected to be expressed in terms of a one-year horizon, which is conventional for most credit risk modelling exercises. Since $\mathbb{E}[\Delta t | S]$ can be significantly smaller than 1, a correction to the LRADR is required. The following transformation is thus required if the duration approach is used to calculate RWA under the IRB approach:

$$LRADR_{S, \Delta t=1} = 1 - e^{\frac{\log(1 - LRADR_{S, \Delta t \leq 1})}{\mathbb{E}[\Delta t | S]}} = 1 - e^{\frac{\log(1 - LRADR_{S, \Delta t \leq 1})}{\frac{1}{N_S} \sum_{i=1}^{N_S} \Delta t_i}}. \quad (14)$$

Here, N_S is the total number of observations across the whole sample for score S . Note that equation (14) is derived from the hazard rate function assuming the hazard rate from time 0 to 1 is constant.

4.3. Constant hazard rate

This approach is similar to the duration approach, but instead of assuming a constant hazard rate for the entire observation period, the hazard rate is assumed constant in time intervals of 0.1. First, the hazard rate is calculated per time interval:

$$\lambda_{S,t} = \frac{-\log\left(1 - \frac{\sum_{i=1}^{N_{S,t}} \Delta t_i D_i}{\sum_{i=1}^{N_{S,t}} \Delta t_i}\right)}{\mathbb{E}[\Delta t | S, t]} = \frac{-\log\left(1 - \frac{\sum_{i=1}^{N_{S,t}} \Delta t_i D_i}{\sum_{i=1}^{N_{S,t}} \Delta t_i}\right)}{\frac{1}{N_{S,t}} \sum_{i=1}^{N_{S,t}} \Delta t_i}. \quad (15)$$

[†] Since in the context of RWA only the PD over a given one-year period is relevant, maturities will be capped at 1.

Table 3. Calibrated coefficients and their standard errors of two Cox regression models and the alternative PBC model on the simulated data.

Variable	Parameter	True value	CPH	TVCPH	PBC
LRADR	γ	0.00181			0.00157 (0.00019)
S	β_1	0.280	0.111 (0.006)	1.400 (0.044)	0.273 (0.009)
Z_t	β_2	-0.300	-0.331 (0.029)	-0.759 (0.045)	-0.754 (0.045)
$S \times Z_t$	β_3			-2.024 (0.132)	

Notes: Standard errors for the PBC estimates are given via the inverse numerical Hessian for β_1 and β_2 , while for γ this is done via bootstrapping and refitting the simulated dataset.

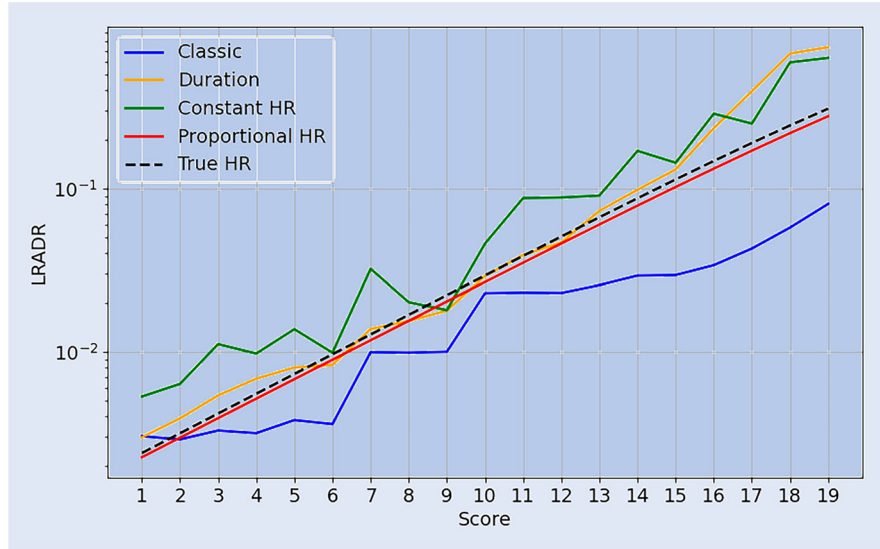


Figure 6. Curves for four different hazard rate approaches to calculating the LRADR on a simulated dataset. The black dashed line is the true hazard rate given the simulation parameters.

Equation (15) is then aggregated per score and transformed to calculate the LRADR.

$$LRADR_S = 1 - e^{-\mathbb{E}[\lambda_{S,t} | S]} = 1 - \exp\left(-\frac{1}{T} \sum_{t=1}^T \lambda_{S,t}\right). \quad (16)$$

4.4. Proportional hazard rate

In this approach, the PBC regression is applied from Section 2 to yield a maximum likelihood estimate for the baseline hazard rate, $\hat{\lambda}_0$,

$$\hat{\lambda}_0 = \hat{\gamma}, \quad (17)$$

where we have set $\Delta t = 1$ for comparability to the other methods. Thereafter the LRADR is calculated by scaling this estimate using the score covariate's estimated slope:

$$LRADR_S = 1 - e^{-\hat{\gamma} e^{\mathbb{E}[\hat{\beta}_1 S + \hat{\beta}_2 Z_t | S]}} = 1 - e^{-\hat{\gamma} e^{\hat{\beta}_1 S}} \quad (18)$$

Note that the term for the economic cycle drops out since the expected value of the systemic factor is 0 (Oeyen and Celis 2019).

4.5. Calibration results

All of the above methods yield a LRADR curve and are implemented for the simulated portfolio. The actual LRADR curve is included, which is functionally equivalent to the Proportional Hazard Rate approach. The estimates used to generate the simulated data are given in Section 2. The results are provided in figure 6. In the next section, we discuss the shortcomings of each method.

5. Discussion

From figure 6, we see that the classic method fails to establish a link between the score and the LRADR, which would lead model developers to the conclusion that the ranking function producing the score is poor. Given that the simulation forces the default risk to be proportional to the score, this would be incorrect.

Both the duration and the hazard rate approach are sensitive to outliers since the constant hazard rate is not a realistic assumption for the simulation presented in Section 2. Specifically, the hazard rate is negatively correlated with Δt . Hence, the assumption that the hazard rate is constant even for smaller time intervals is still incorrect.

The proportional hazard rate, which leverages on the alternative PBC regression's maximum likelihood estimates, is able to replicate the LRADR used for the simulation with a small margin of error. It should be noted, however, that there is no guarantee that the true default process follows a proportional hazard rate model.

With the results shown in the previous section, it is demonstrated that all methods, except the classic approach, can be used to properly evaluate whether the ranking function is able to properly discriminate default risk. With respect to the method that is most reliable, it is the opinion of the authors that the assumptions of the proportional hazard rate model will pass statistical tests more easily in practice. This is mainly due to the assumptions introduced in the duration and constant hazard rate approaches.

6. Conclusion

Performing a Cox regression when time is negatively correlated with one of the explanatory variables results in an incorrect estimate of the baseline hazard, as well as the slope of the negatively correlated covariate. If the baseline hazard and the slope of the covariate are intricately linked, then neither one nor the other can be approximated analytically by performing a greedy search. In this paper, a rigorous framework is introduced on how to estimate a Cox regression given a parametric baseline hazard, i.e. an exponential duration model. This is done via a bounded maximum likelihood function for a set of simulated Bernoulli trials. The upper bound of the baseline hazard uses the properties of the proportional hazard rate model to ensure that its estimate remains within its feasible domain. This upper bound is set by means of calculating an empirical hazard rate for a subset of the population, namely, the empirical default rate for observations with the best scores and loan durations of one year.

Such an issue has not yet been addressed in the survival analysis literature, since it is an unfamiliar problem within the field of medical research and other popular areas where survival analyses are often used. Notwithstanding, it is argued in this paper that such a problem is quite common for certain type of credit portfolios, and should therefore be considered when developing PD models used to assess the credit risk of a given portfolio. To exemplify this, default curves of a real-world portfolio at ING provides more insights into the problem and its implications for default risk quantification. It is the opinion of the authors that this perspective can have a significant impact on how survival analysis is used by practitioners in the banking industry. It is expected to allow for new types of data and therefore open up new research for Cox regressions in general.

The preferred approach by the authors is the PBC regression introduced in this paper. This approach explicitly parameterises the baseline hazard function in a proportional hazard model, which is done with an exponential duration model and one or more correlated covariates with the duration. Three alternatives that approximate the proposed proportional hazard rate model are applied and compared to the PBC regression. However, it is argued that this leads to less reliable results due to assumptions that will likely not hold in most

cases. Furthermore, the alternative approaches increase the data requirements with respect to the number of observations, which is an undesirable situation in case of sparse data sets.

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Data availability statement

The data that support the findings of this study are protected, confidential, and owned by ING Bank. The data source itself is ING's internal data warehouse, which is a secured repository that is not accessible to the public for privacy, security, and regulatory reasons. Anonymised data can be made available upon reasonable request from the corresponding author, Themis Rallis, provided that those requesting the data sign confidentiality forms/NDAs from ING.

Disclosure statement

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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